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Institutional Disruptions and the Philanthropy of Multinational Firms

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Abstract. This paper studies philanthropy by multinational enterprises (MNEs) during institutional disruptions—the sudden and unexpected, temporary, and systemic breakdowns in market-oriented institutions. The central argument is that, under institutional disruptions, MNEs aim to restore factors that are essential for the market to function, such as infrastructure and labor markets, and the strength of the market restoration motive is positively associated with the economic importance of the affected country to the MNE. Analyses of donations from 2,000 MNEs headquartered in 63 countries in the aftermath of 265 major epidemics, natural disasters, and terrorist attacks affecting 129 countries suggest that the economic importance of the country to the firm strongly explains donations. Country market concentration, public aid, and the country's regulatory quality moderate this effect. These associations are robust to a matching method; a vector of firm-, country-, and event-specific time-varying and -constant variables; and alternative motives, such as reputation, altruism, media salience, market standing, and poverty-gap avoidance. They offer evidence that company philanthropy in the aftermath of institutional disruptions may deviate from predicted behavior under stable conditions. Particularly, the findings contest the expectation that philanthropy rises in market competition. Monopolistic firms are comparatively large donors and may act as an economic stop-loss mechanism during large disruptions.

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Keywords: company philanthropy • corporate social responsibility • disasters • institutional disruptions • multinational enterprises

Introduction

Over the past decade, substantial research in the field of organization science has explored the motivations underlying for-profit organizations' involvement in addressing large societal issues (George et al. 2016, Cabral et al. 2019, Luo and Kaul 2019). An extensive literature studies company philanthropy toward chronic societal problems under stable institutional contexts (e.g., poverty reduction and inadequate education and healthcare) (Bénabou and Tirole 2010, Lev et al. 2010). Yet an increasingly important societal issue stems from what we term *institutional disruptions*—the severe and systemic, yet sudden, unpredictable, and temporary breakdown of market-oriented institutions. These institutions are the intangible and physical factors necessary for business activities (Coase 1988, Acemoglu et al. 2005, Chakrabarti et al. 2011, Dutt et al. 2016). Phenomena such as epidemics, natural disasters, and terrorist attacks disrupt country market-oriented institutions. They halt firms' ability to obtain inputs or sell because of the destruction of infrastructure and purchasing power (Becerra et al. 2014, Boehm et al. 2019); reduce productivity because employees lose housing, schooling, or health status (Altay and Ramirez 2010, Baker and Bloom 2020); lead to suspended regulatory and enforcement systems (Copeland 2005, Garrett and Tetlow 2006); and cancel investments because of the impact on financial intermediaries (Hosono et al. 2016).

Since the turn of the 21st century, multinational enterprises (MNEs) have become the fastest growing sector worldwide in funding recovery from disruptions, outpacing governments, multilateral agencies, and individual charity (High-Level Panel on Humanitarian

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Financing 2016). Considering that these shocks can create poverty traps and hamper economic development (Barro 2009, Baker and Bloom 2020), understanding MNE motivations to donate in the aftermath of institutional disruptions is crucial for fostering and coordinating important resources for country recovery.

Scholars thus far have applied motives widely tested under stable institutional contexts to explain philanthropy after such events. That is, firms donate to meet social preferences, such as altruism or reciprocity (Morgan and Tumlinson 2019), or pursue strategic considerations, such as to accumulate reputational capital (Muller and Kräussl 2011), hedge the firm against political risk (Zhang et al. 2016), or comply with social norms or political pressure (Tilcsik and Marquis 2013, Zhang and Luo 2013, Luo et al. 2016). These approaches have deepened our understanding of company philanthropy. In this study, and by building on the importance of the firm's locations with an economic rationale, we suggest that institutional disruptions provide a context that makes salient a different, previously not studied motive: restoring the market.

We theorize that institutional disruptions reveal the strategic value of a country's market for firms. Company performance relies on the market-oriented institutions in the countries in which the firm operates (Khanna and Palepu 1997, Toulan 2002, Chittoor et al. 2009), and disruptions increasingly explain performance volatility and survival for firms with operations in the affected country (Baker and Bloom 2020, Aghion et al. 2021). The larger the economic importance of a country for the MNE (i.e., the extent to which the firm sells, buys, or rents inputs, final products, or services or hires human capital in the country), the greater the shock to firm performance and, thus, its incentive to allocate resources toward the country's market recovery.[1] Consequently, a measure of the relative economic importance of a country market for MNEs can identify which firms will donate when disruption strikes.

Drawing on arguments on the costs and benefits of nonmarket decisions and our interviews with business decision makers from S&P 500 firms,[2] we argue that this baseline relationship is moderated by the availability of alternative sources to fund market recovery and the nature of the market institutions (Buchanan 1965, Yildirim 2014, Kaul and Luo 2017). Along with the formal insurance system, countries use two alternative sources for funding recovery: private and public aid. Thus, we focus on other corporate donors and aid coming from the state power and foreign agencies as substitutes for a focal firm's giving.

First, when the pool of large company donors is small, the cost of foregoing nonmarket action is considerable. We hypothesize that the average donation of an economically connected firm falls in the population of MNEs with a large share of the country market. Second, a country's recent history of substantial public expenditure and foreign aid fosters the expectation that government resources will fund recovery and, thus, will be associated with smaller donation amounts from economically connected MNEs. Finally, low regulatory quality raises the cost of funding recovery because of bureaucratic delays and the misallocation of resources (Becerra et al. 2015, Ballesteros and Gatignon 2019). Accordingly, we expect economically connected MNE donations to increase with country regulatory quality.

To empirically test the hypotheses, we construct the Global Database of Disaster Responses. The database covers reported monetary and in-kind donations from firms, governments, multinational agencies, and nongovernmental organizations reported in news media to the relief and recovery of all disasters (epidemics, natural disasters, and terrorist attacks) from 1990 to 2019. We select shocks that meet the characteristics of institutional disruptions to our event panel. Our firm panel is the 2,000 largest public MNEs (by revenue) at the international level and their affiliates worldwide. We construct a time-varying, pair-specific measure linking ownership-weighted company operations to countries for the relative economic importance of markets for each MNE.

Consistent with the strategic motive of donating to restore the country market, the analyses provide systematic and robust empirical evidence that the economic importance of a country to the MNE is positively associated with its philanthropic response in the aftermath of institutional disruptions. The association is significantly moderated by the theorized constructs that influence the costs and benefits of donating for country market recovery. Moreover, we exploit variation in donation reports using natural language processing to identify the allocation of cash and in-kind resources toward factors critical for the functioning of market-oriented institutions. Our data indicate that economically connected MNEs frequently donate toward such areas, which gives additional evidence of the argued philanthropic motive.

The results survive the inclusion of a battery of fixed effects and time-variant firm-, country-, and event-specific variables, including the standard measures of other major alternative explanations of company philanthropy, such as reputational capital, pressure by the host country government, market growth, media visibility, inequity and poverty aversion, and the altruistic motivations of employees. We also apply a matching technique to produce estimates that de facto condition

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on the possibility of self-selection of MNEs into countries with specific exposure to disruptions or proneness to receive aid and find consistent results.

We contribute to the nonmarket literature by bringing attention to a different underlying motive of company philanthropy. A market restoration motive leads to predictions that diverge from extant work. Specifically, research suggests that organizations with operations in an affected country are likely to incur substantial damages and, therefore, will engage in scarce giving in the aftermath of large shocks (Crampton and Patten 2008, Tilcsik and Marquis 2013). In contrast, using a comprehensive database across disruptions, countries, and time, we find systematic evidence that economically connected MNEs are the largest donors in the context of institutional disruptions. When market-oriented institutions are disrupted, limiting resources to the organization's assets can be insufficient for reinstating performance.

Additionally, scholars predict that the quest and potential returns to differentiation via prosocial behavior are large in competitive markets (Kitzmueller and Shimshack 2012), that company giving aligns with pledges from public actors because of the interest in using political capital as a risk-management mechanism against stakeholder opportunistic behavior (Henisz et al. 2014, Bertrand et al. 2020), and that such incentives rise in institutional underdevelopment (Zhang and Luo 2013, Hornstein and Zhao 2018). In contrast, our findings indicate that companies donate comparatively large amounts in highly concentrated country markets with high regulatory quality, and their giving misaligns with that of public entities.

The study also contributes by distinguishing the concept of institutional disruptions and its relevance to company behavior. The concept may help to delineate the economic implications of these phenomena on company performance and the possible strategies firms may use to manage them. Institutional disruptions are ubiquitous to emerging and developed economies alike and are distinct from the long-lasting or permanent absence of market-based institutions (voids) in emerging market countries (Khanna and Palepu 1997, Doh et al. 2017). The results consistently indicate that this theoretical distinction is meaningful. For instance, the overlap between private and public benefits in concentrated country markets appears to be substantially greater during institutional disruptions. Monopolistic firms seem to act as a stop-loss mechanism when disruptions overwhelm the capacity of governments. We observe this across levels of economic and institutional development.

Finally, we contribute to the literature on shocks by formally characterizing them based on temporality, suddenness and unexpectedness, severity, and pervasiveness. Existing research considers one or several of the characteristics but does not formalize all four.[3] For instance, studies focus on expectedness and severity (Dye et al. 2014); severity, frequency, and expectedness (Oetzel and Oh 2014, 2021); and severity, pervasiveness, and permanence (Klüppel et al. 2018). Our framework allows for the combined study of different human and naturally caused events, such as terrorist attacks, natural disasters, and epidemics. It enables us to differentiate which shocks have disruptive rather than evolutionary institutional impacts, allowing testable predictions on firm outcomes and behavior. This study identifies how the economic importance of the country to the MNE and country conditions are associated with MNEs' response to large-magnitude shocks.

The different main and robustness analyses highlight the conditions and mechanism under which the motive to restore the country market is likely to be active. In this way, this study complements research on the economic efficiency of the company behavior that indicates firm giving speeds economic recovery (Ballesteros et al. 2017). By helping restore market-oriented institutions, MNEs become key actors to help overcome societal issues around the world.

Theory and Hypotheses

Market-Oriented Institutions and Firm Operations

Markets are "institutions that facilitate the exchange of goods" (Coase 1988, p. 7). Market-oriented institutions combine to create interlocking activities that shape business in the country market (Nelson 1995). They entail not only product, capital, and labor markets, but also intermediaries, physical infrastructure, and regulatory and enforcement systems (Coase 1988, Khanna and Palepu 1997, Acemoglu et al. 2005, Chan et al. 2008, Chakrabarti et al. 2011, Dutt et al. 2016). Market-oriented institutions affect firm costs of exchange and production and the ability to transact or operate. Underpinning the functioning of market-oriented institutions is the presence of tangible and intangible factors that allow market transactions to occur. When these factors disappear or become damaged or destroyed, there is a breakdown in market-oriented institutions.

Traditionally, researchers study the absence or evolution of market-oriented institutions with two broad streams of scholarship. On the one hand, the chronic absence of goods for country market operation, referred to as *institutional voids*, creates market failures (Khanna and Palepu 1997). When factors such as communication, transport, energy infrastructure, skilled labor, intermediaries, and contract enforcement mechanisms are absent, firms face substantial operational and transactional frictions (Khanna and Palepu 1997, Chan et al. 2008, Doh et al. 2017). These frictions

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impede firms from accessing the resources and agents necessary for economic activities and have significant organizational and performance implications. Consequently, firms with the relevant resources often internalize activities, and those without such resources may not be able or may choose to not operate in the country market.

On the other hand, scholarship on evolutionary changes to market-oriented institutions studies permanent alterations. The institutional changes are mainly a result of slowly evolving socioeconomic and political processes but may (more rarely) be punctuated changes (Newman 2000, Peng 2003, Klüppel et al. 2018). Punctuated changes include upheavals (Newman 2000), transitions (Peng 2003, Chittoor et al. 2009), and traumatic shocks (Klüppel et al. 2018).[4] Institutional upheavals and transitions are rapid and comprehensive changes that enduringly alter the systems and underlying factors for the competitive landscape of a country's market (Newman 2000, Peng 2003). Traumatic shocks, such as war, change institutions to affect firm strategy and structure decades or even centuries afterward (Klüppel et al. 2018). The radical changes lead firms to adjust their operations to fit the new institutional environment and survive.

The literature is not explicit about institutional disruptions, which we posit are different from existing concepts of institutional voids and evolutionary changes in four cardinal ways.

Short-Lived Temporality. In contrast to the persistent shifts from evolutionary changes and chronic absence from voids, disruptions are temporary deviations from the status quo. Although there may be some adjustments in norms, such as building codes, most institutions return to a similar level after an institutional disruption (Useem et al. 2015, Bloom et al. 2018).

The transient nature is driven by the phenomena behind institutional disruptions developing and disappearing in days yet leaving a wake of destruction whose length of restoration may be substantial. For essential goods, such as education and housing, repair often comes within weeks (Tomasini and Van Wassenhove 2009). For others, such as transportation infrastructure, rebuilding may take several years (Ballesteros and Kunreuther 2018). For instance, it took nearly three months to restore the water system in Japan after the Tōhoku earthquake (Ballesteros 2019). The displacement and malfunctioning of labor markets after U.S. Hurricane Katrina lasted more than two years (Deryugina et al. 2018).

Suddenness and Unexpectedness. For institutional voids or evolutionary changes, checks and balances and due processes using macro variables facilitate assessing challenges from the lack of or underdeveloped market-oriented institutions (Henisz 2000). Decision makers can assess country attractiveness (Berry et al. 2010) and adjust their organizations' structures and strategies (Toulan 2002, Chittoor et al. 2009) to the evolving institutional context (Flores and Aguilera 2007, Wu and Chang 2014). For institutional upheavals, transitions, and traumatic events, the changes typically take two to three years to unfold (Newman 2000), providing firms time to adjust their operations.

Conversely, institutional disruptions are highly unpredictable and characterized by rapid drops in the functioning or availability of factors necessary for market operation. In this sense, traditional measures of country risk and development are often poor indicators of the resilience of economic institutions against disruptions (Ballesteros et al. 2017). To illustrate, before the Tōhoku disaster that resulted in the largest economic damage from one event in history, scientists calculated a zero probability that a hundred-foot tsunami could hit Japan (Ferris and Solis 2013), and many firms believed that the Japanese economy barely would be affected (Kunreuther and Useem 2018). The CEO of a chemical company explained to us, "We got together all of our leaders and we had them work through three different scenarios using data from the government on what they would do … we had those plans in place … [The loss] turned out to be a lot worse."

High Severity. Whereas evolutionary changes often target improved market-oriented institutions, disruptions entail significant destruction in the affected country. For instance, in the United States, the Federal Emergency Management Agency calculates that 40% of businesses do not reopen after experiencing a natural disaster, and 90% go bankrupt within a year if they do not resume operations in a week (Federal Emergency Management Agency 2015). The disruptions result in an extensive loss in connectivity with stakeholders (Alfaro and Chen 2012) and, more broadly, the provision of products and services (Bloom et al. 2018). The 2010 Chile and 2011 Japan earthquakes and tsunamis, for instance, destroyed, respectively, 32% and 17% of goods supply for at least two months (Cavallo et al. 2014).

There is systematic evidence that these events explain economic development, and one key mechanism through which this occurs is drops in productivity (Bloom 2009, Altay and Ramirez 2010). In practical terms, organizations face reductions in the ability of their labor force to work at the same level. Not only are employees or their family members often direct victims and their residencies affected, which implies that they must relocate, but they may also suffer the loss of their children's school or the transportation systems to commute. Crucially, this affects employees

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of both business and government organizations. As a municipal official commented on the 2010 Chile earthquake, "There was no light, we had no water, no staff, the city disappeared, officials disappeared … There was absolutely no one." The magnitude of the consequences was ubiquitous in our interviews. After the Japanese 2011 earthquake and tsunami, a manager of a logistics company stated, "[The firm is] worried about roads, trains, airports. Things that get produced are going to be delayed and also the materials to produce them. Everyone is going to miss their numbers."

Pervasiveness. Scholars traditionally situate institutional voids in lower-income countries or emerging economies (Khanna and Palepu 1997, Doh et al. 2017) and evolutionary changes in transitional economies (Toulan 2002, Peng 2003, Banalieva et al. 2015). Conversely, institutional disruptions spread across economic and institutional development levels with costlier shocks affecting medium- and high-income countries (High-Level Panel on Humanitarian Financing 2016). MNEs increasingly acknowledge this ubiquity in our interviews. The head of operational risks of a bank indicated that "before the terrorist attacks [of 9/11 in the United States] there was a sense that things happen in places like Nigeria, but they don't happen in places like New York City."

The effects of institutional disruptions are systemic at the country level. Interdependencies transmit shocks across industries and country regions. For example, Thailand, which manufactures one third of the world's hard-disk drives and is a production hub for carmakers, experienced a flood in 2011 that damaged or destroyed the facilities of more than 14,000 businesses. Companies such as Apple, Toyota, and Ford that relied on operations or suppliers in the country had to suspend or delay production because of part shortages and the inability to access alternative sources of parts. The effect on financial intermediaries is similarly problematic. To illustrate, after the Tōhoku earthquake, 11.4% of firms operating in Japan indicated that their bank could not operate (Miyakawa and Hosono 2014), causing otherwise undamaged firms to have significantly smaller investments (Hosono et al. 2016).

Donations as a Response to Institutional Disruptions

Significant changes in market-oriented institutions affect firm performance and survival, motivating their response (Toulan 2002, Chittoor et al. 2009). Firms can react by altering their governance structures to fit the institutional environment or by transforming the institutional environment (Williamson 1998, Dorobantu et al. 2017). A vast literature focuses on firms matching their governance structures to the institutional environment (Williamson 1991, 1998). Nevertheless, institutional economists have long acknowledged that firms should first get the institutional environment right (first order economizing) and then select the best governance structure based on the institutions (second order economizing) (Williamson 1998). As discussed, the characteristics of disruptions are crucial to understanding MNE response.

More specifically, the temporal nature of disruptions attenuates incentives to change the supply chain, alter governance structures, or exit the country market. Organizations benefit from close proximity, interfirm asset specialization, and relationships developed over time with their exchange partners (Dyer 1996, Elfenbein nd Zenger 2014). Switching suppliers can be expensive and create a loss of valuable relational capital (Elfenbein and Zenger 2014). Additionally, exiting the country market often is not an attractive option given the access to customers, resources, or local knowledge (Alcácer and Chung 2007). Consequently, these are often costly or undesirable strategic responses to disruptions.

The abruptness and ambiguity of disruptions often yield prevention and coping mechanisms unfeasible or ineffective. For instance, despite more than 75% of the firms that we interviewed indicating that they had routines for threat identification and prioritization against disruptions, they note that the relevant required information for planning is mostly unavailable and inaccurate. Moreover, although, over the last 20 years, firms have had sophisticated instruments to hedge catastrophe losses (e.g., index-based insurance, such as catastrophe bonds), the financial impact of these shocks has skyrocketed, resulting in an uncovered magnitude of hazard (Kunreuther and Useem 2018).

The multifaceted and pervasive nature of disruptions means that firms often need to consider strategic actions beyond their operations. Additionally, their high-magnitude consequences incentivize decision makers to allocate resources that mitigate impacts on firm performance. Although firms may lobby governments to distribute public aid for country market recovery, such indirect strategies unfold slowly (Cohen and Werker 2008) and can be costly as the firm waits for restoration.[5] Donations can be a direct and effective means of transforming a disrupted institutional context.

Our database provides evidence of the market restoration motive for philanthropy. In response to the 2011 Thailand flood, Honda donated $3.7 million and 200 generators for the Thai government to provide power in areas so that customers, businesses, and government organizations could function. Another case is when Hurricane Katrina severely damaged Royal Dutch Shell's large operations in the Gulf of Mexico.

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The MNE helped rebuild public facilities and assist agencies, including the New Orleans Police and Justice Foundation's housing initiative for police, fire, and emergency medical staff. Additionally, Shell provided $45 million to support its employees and "made getting them back to work a priority" (Ballesteros 2019).

At the same time, given the uncertainty of institutional disruptions, this type of philanthropy is rarely budgeted. The firm must often face the opportunity cost of deferring country investments or cancelling previously planned social projects (Ballesteros et al. 2018). Corporations regularly reallocate resources from previously approved causes that account for their annual social responsibility budget or even more. Take the case of Pfizer, which donated $47 million in the aftermath of the 2004 Indian Ocean tsunami or many times its combined annual social expenditure in the eight affected countries.

Given the opposing forces of the need to recover the country market and the opportunity costs of philanthropy, a central question is what affects when an MNE becomes a contributor toward recovery. Donating to restore the elements necessary for market-oriented institutions to function is a nonmarket strategy that helps to reestablish the institutional context. Firms are motivated to influence their institutional environment based on the costs they face and their perceived benefits from the nonmarket strategy. In the context of disruptions, this logic allows predicting which MNEs will donate.

The Economic Importance of Countries and the Philanthropy of MNEs After Disruptions

MNEs with a large proportion of their performance explained by the affected country suffer greater financial consequences than similar firms without such a connection (Alfaro and Chen 2012, Aghion et al. 2021). One might expect firms with economically important operations to focus their resources on restoring their facilities and engage in scarce giving (Tilcsik and Marquis 2013). Economic concerns constrain company philanthropy, and the disruption creates fiscal pressure on the firm. Additionally, affected nonprofits create a broken country infrastructure for philanthropic giving (Ballesteros and Gatignon 2019).[6] Altogether, these factors suggest that MNEs with economically important operations donate less after an institutional disruption.

Contrary to this prospect, we argue that these firms will be the *largest* donors for several reasons. First, institutional disruptions have a systemic impact. Thus, although a firm may rebuild its operations, the lack of factors necessary for market-oriented institutions to function means that it still may not achieve predisruption performance levels until market institutions are reestablished. For instance, the 2011 Tōhoku disaster extensively damaged three Toyota factories, destroyed Japanese ports, and crippled the power supply. Consequently, even if Toyota had rebuilt its plants, it would not have been able to operate fully. Its suppliers could not supply components without power, nor could it ship its automobiles or supply auto parts to its international manufacturing facilities until the ports were restored (Forbes 2011). Toyota donated power supply systems to ease power shortages in the country and paid full salaries to their employees to help its suppliers start producing again (Rodríguez and Sanchez 2012).

Second, firms are motivated to provide financial resources to minimize the spillover effects across affiliates (Gopalan et al. 2007) and are more likely to respond where the impact of existing institutions on the firm's business is greater (Hillman et al. 2004, Hadani and Schuler 2013). For firms whose performance is largely explained by the affected country market, the costs of not restoring the country market are extensive. Affiliates of strategic importance are likely to receive support from top management in response to institutional changes because the impact is material to the firm (Gubbi et al. 2015). Our interviewees frequently refer to this idea. A manager of a manufacturing firm commented, "The situation in Thailand was very different from Japan following the earthquake in an important way. When the impact [hit] a size of our business that was much greater, we immediately mobilized resources … We knew that if it escalated, it will be bad for us because of a larger supply chain disruption … in this case, a component sourcing from Thailand." Likewise, the vice president of a British mining company remarked on the 2010 Chilean earthquake: "Chile is [our] number one copper producer, and that represents a big part of our business … so we mobilized … "

Third, this type of philanthropy has immediate financial payoffs. MNEs with large host country operations are more efficient at identifying areas instrumental for country market functioning, providing capital faster than traditional sources of aid and assisting rebuilding efforts toward recovery, which reduces the cost of philanthropy (Ballesteros et al. 2017, Kunreuther and Useem 2018). For example, the mining company mentioned earlier used its technology and machinery for restoration, commenting, "The first action, to clean the roads … and allow communication and transportation systems to be restored." By taking these actions, not only could the firm start transacting again, but also its customers and suppliers could operate, and thus, it helped reinstate the country market system. Consequently, the benefits extend beyond the focal donation and are proportional to the economic importance of the country to the MNE.

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Hypothesis 1. *The greater the economic importance of a country for the firm, the greater its donation in response to an institutional disruption.*

As previously discussed, we expect that a country's concentration of dominant MNEs, public aid, and regulatory quality are three contextual elements that moderate the baseline relationship.

Dominant MNE Index. A firm's share of benefits and costs from influencing market-oriented institutions depends not only on where the business impact is greatest, but also on whether other firms are able to participate. A country's index of dominant MNEs reflects the concentration of large firms in the country. When the country is highly competitive—that is, when there are a large number of players with sizeable country market shares—then the costs of changing the institutional environment may be divided across more firms capable of contributing (Sandler 2013). This tempers economically connected MNEs' incentives to donate.

In practical terms, MNE managers that we interviewed discussed how, in deciding how much to donate in an economically important country, they look to the presence of other large firms in the country such as Coca Cola, Microsoft, Mitsubishi, and Tata. Ceteris paribus, the fewer firms holding a significant share of the country market, the greater the strategic value of restoration for MNEs with economically important operations. Monopolistic firms realize comparatively large proportions of the benefits from restoring market-oriented institutions (Bonardi et al. 2005, Luo and Kaul 2019).

After the 2010 Chile earthquake, managers from a large mining company reflected that its operations accounted for approximately 3% of Chile's GDP to emphasize the marginal relevance of operative interruptions. The firm donated $10 million toward rebuilding. For monopolistic firms, the cost-benefit of giving approximates private investments. These firms can capture comparatively large value from the capital injected to restore the country market.

Hypothesis 2. *The dominant MNE index positively moderates the relationship between the economic importance of a country to the MNE and its donations in response to an institutional disruption.*

Public Aid. Public actors traditionally play an essential role in managing societal challenges. For disruptions, public relief and recovery come from the host country and foreign governments and multilateral agencies, such as the United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA) (High-Level Panel on Humanitarian Financing 2016).[7] Thus, host country public aid and foreign public aid are two distinct sources of aid that MNEs consider in donating. A recent history of substantial host country and foreign public aid to restore human welfare, damaged infrastructure, and other factors necessary for market-oriented institutions fosters an expectation by firms that public aid will fund the country's recovery.

There is evidence at the individual level that public intervention crowds out philanthropy (Andreoni 2006, Yildirim 2014). We hypothesize that a similar relationship operates at the organizational level because public aid brings the benefits of country market restoration to the MNE without incurring the costs of giving. Conversely, when host country and foreign public agencies do not mobilize resources in sufficient magnitude, MNEs with high economic importance are more likely to deem firm resources as costs critical for achieving country market restoration.

This dynamic is captured by a vice president of a soft drink company when explaining his firm's philanthropic engagement after the 2011 disaster in Japan: "We are part of a system. If the [Japanese] government cannot [effect a recovery], we need to rebuild. We need the market to recover." MNEs with high economic importance in the affected country donate more after a disruption when they believe that its economic magnitude outstrips public resources.

Hypothesis 3a. *Host-country public aid negatively moderates the relationship between the economic importance of a country to the MNE and its donations in response to an institutional disruption.*

Hypothesis 3b. *Foreign public aid negatively moderates the relationship between the economic importance of a country to the MNE and its donations in response to an institutional disruption.*

Regulatory Quality. A country's governance quality influences the materialization of incentives to engage in nonmarket behavior (Aguilera et al. 2007, Zhang et al. 2016, Dorobantu et al. 2017). Government commitment and ability to formulate and implement policies to permit and promote private sector development, referred to as regulatory quality (Kaufmann et al. 2011), are critical for creating and maintaining market-oriented institutions. High regulatory quality enables the swift use and dedicated allocation of resources toward factors necessary for economic activity. It, therefore, enhances the comparative benefits of firm donations toward the recovery of country markets. On the other hand, low regulatory quality increases the costs of donating. In such case, the process of rebuilding can be substantially impeded by lower government commitment, ability, or red tape (Becerra et al. 2014). Even if the MNE uses its efforts and resources to rebuild directly, the logistical complexity from regulations

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and controls reduces the efficiency of such efforts (Ballesteros and Gatignon 2019).

Some of our interviewees note these problems after the Chile earthquake in 2010 when companies were encouraged to contribute to recovery through the government's ministries. Firms soon realized that government inefficiencies mired the delivery of funds. As one manager stated, "We were worried that the public system would take forever to apply resources to the emergency … and that, for sure, led firms to think twice about donating."

Donations are more likely to be diverted or suffer leakages before reaching the target areas when regulatory quality is low. For example, reports suggest that the government's misuse of aid may be behind the lengthy recovery of Haiti from the 2010 earthquake (Useem et al. 2015). Country officials demanded cuts of donations and held up goods at borders to force payment (Farny et al. 2019).

Thus, low regulatory quality deters donations from economically connected MNEs because decision makers fear that resources will be delayed or not well spent, which increases the relative cost of giving. In such environments, skepticism abounds as to whether the benefits of donations will materialize as one survivor of the Haiti earthquake commented, "There might be some more money [from the donors], but those who need it won't receive it" (Fletcher and Guyler Delva 2010, p. 1).

Hypothesis 4. *Country regulatory quality positively moderates the relationship between the economic importance of a country to the MNE and its donations in response to an institutional disruption.*

Empirical Strategy

Data

Company Philanthropy. We built the Global Database of Disaster Responses using automated Boolean searches and natural language processing in Python. The database contains information on reported cash and in-kind donations from organizations (firms, governments, multinational agencies, and nongovernmental actors) for the relief and recovery of every major disaster that affected the world from 1990 to 2019. The Boolean searches in Factiva, Google, and Lexis Nexis covered newspapers, trade press, magazines, newswires, press releases, TV and radio transcripts, digital video and audio clips, corporate websites and reports, institutional websites and reports, and government websites and reports, among other sources and resulted in more than 2,310,000 reports. We made the reports computationally tractable with differential language analysis and code information on the donor organization, the characteristics of the donation (i.e., in-kind, monetary or both, amount, currency, and timing), the initiator within the firm (i.e., employees or top management), and the target of the donation (i.e., essential goods, factors for the market, etc.). The database covers 96,858 donations from 40,170 firms from 84 countries of origin.

The online appendix (https://institutionaldisruptions.com/) contains the detailed procedures for data collection and coding, including the monetization of in-kind donations and conversion of non-U.S. dollar reports.[8] It also describes the independent assessments of data quality and checks with third-party sources, including UNOCHA.

Firm Panel. Internationally large corporations account for more than 90% of the growth in the participation of firms on disaster aid (Ballesteros et al. 2017, Kunreuther and Useem 2018). We rely on the largest 2,000 publicly traded MNEs by total revenue at the international level from Orbis to cover a representative sample of these firms. The MNEs had a total of 545,664 affiliates over the 2007–2019 period.

A major task was cleaning the ownership of the MNE affiliates for entry and exit into the MNE resulting from mergers, acquisitions, dissolutions, and spinoffs. We use shareholder and legal event data from the Orbis database to code changes in ownership over time. When data were missing or conflicting information was provided in the database, we cross-checked the data with public filings, corporate websites, and internet searches. Procedural details are in the online appendix.

Country and Disruption Panels. Data on public aid, regulatory quality, and country-level controls came from the World Bank's World Development Indicators (WDI) and the Worldwide Governance Indicators (WGI) databases.[9] We gathered data of all shocks from the International Disaster Database (EM-DAT) created by a center supported by the World Health Organization.[10] For each shock, we obtained data on human and economic loss from the UNOCHA and the reinsurance company Swiss Re.[11]

Measures

Institutional Disruptions. For institutional disruptions, we identify sudden and unpredictable shocks that created systemic and severe losses in the country market. First, we follow Baker and Bloom (2020), who show that large terrorist attacks and natural catastrophes are associated with abnormal levels of market volatility and significantly explain country GDP growth. Barro (2009) also includes epidemics as phenomena with several times more welfare costs for countries than frequent economic fluctuations. Additionally, large epidemics, natural disasters, and terrorist attacks tend to have systemic impacts on the country market

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(Aghion et al. 2021, Bloom et al. 2018, Kozeniauskas et al. 2018). We then adopt the 30-day criterion between the start and end dates consistent with Ballesteros et al. (2017) for sudden shocks.

Finally, to characterize severity, we calculate the percentile distribution for deaths, affected people, and economic damage. The distribution of each variable was calculated by country for all shocks reported between 1997 and 2019 in EM-DAT, which uses a minimum degree of impacts to record events. We use percentiles because the mean and standard deviations are inefficient location statistics given the skewness of the historic impact distribution. We choose shocks that rank at the 99th percentile in the affected country in any of the three impact areas (i.e., *deaths*, *affected people*, or *economic damage*; Cavallo et al. 2013) to capture severe shocks for the country. We test the robustness of the results to the 75th and 90th percentiles (Table 4 in the Online Appendix).

The firm-level data covers 2007–2019 that had 4,273 shocks worldwide. From these, 3,822 shocks passed the 30-day criterion. At the 99th percentile of severity, there are 265 shocks that affected 129 countries across the four levels of country income.[12] Procedural details of the identification of disruptions is found in the Online Appendix, which also presents the severity cutoffs by country (Table 1), the country-income classification thresholds (Table 2), and the list of institutional disruptions (Table 3).

Dependent Variable. The dependent variable *Donation* is a continuous variable of the total U.S. dollar amount donated, logged for the analyses. We further refined the internal validity of our measure with *Donations Toward Market Factors.* We calculate it by summing donations whose reports explicitly indicate targeting the factors underpinning the functioning of market-oriented institutions. The factors include rebuilding transportation, communication, power, and other infrastructure, and goods necessary for labor country market functioning, such as housing for displaced workers (see online appendix for details). We also use this variable to test alternative explanations, such as the pursuit of reputation. We, however, do not use it as our primary variable because companies self-select to report the target of the donations.

Economic Importance. Measuring the economic importance of a country to the MNE is not trivial given affiliates' idiosyncratic characteristics that affect their strategic value and data constraints. For instance, measures based on revenue may strongly represent sales and underrepresent production affiliates. In contrast, measures based on employees and assets may underrepresent the strategic value of highly automated affiliates that employ few people. These measures largely reflect the demand side of the market rather than both the supply and demand side. Although industry and country-level fixed effects can mitigate some of these concerns, there are substantial missing data for affiliate-level revenues, employees, and assets in databases. Although there is greater database coverage of the existence of affiliates than of their financial and operational metrics, a measure of economic importance based on the proportion of MNE affiliates in a country assumes an equal strategic value for each affiliate.

To operationalize a country's economic importance to the MNE, we start with the proportion of total *affiliates* owned by the MNE in the affected country. The international business literature has consistently identified the geographical dispersion of affiliates as a determinant of corporate performance (Flores and Aguilera 2007, Chakrabarti et al. 2011, Magelssen 2020). We adjust the measure by the MNE's *ownership* percentage of each affiliate because the strategic value of country-specific performance for the firm rises in ownership. For countries where the MNE has no operations, economic importance equals zero. *Economic importance*, **k**, of country **c** to firm **f** in the year of the disruption **t**, ranges from zero (low) to one (high) and has the following form:

$$k_{ct}^{f} = f(\sigma_c^f)_t, \tag{1}$$

where $\sigma_c^f = \frac{\sum_0^n (affiliate * \%\ ownership)}{\sum_0^w (affiliate * \%\ ownership)}$; n is the number of firm affiliates in country c and w is the number of firm affiliates in the world in the year of the disruption.

We test alternative explanatory variables with individual and combined polynomial expansions for affiliate *revenue*, *assets*, and *employees* (Table 4). The tests consistently show that a measure based on affiliates has more explanatory power and it is a less biased estimator than variables of the financial book. These variables can reduce construct validity. For instance, revenue, number of employees, and capital intensity can overrepresent some industries and some types of affiliates, such as sales subsidiaries. They may only represent the demand or supply side of the market or may not capture the strategic value of an affiliate as a source of key resources for the company. Additionally, we run models excluding headquarters (Online Appendix Table 6) and a measure not adjusting for ownership (omitted). The results are qualitatively consistent across these alternative measures.

We analyze missing data patterns and find no systematic divergence between those with missing data and those with available data. In other words, the hypothesis that data are missing completely at random cannot be rejected. To deal with missing data, a polynomial expansion can approximate the complete data.

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We use a multiple-input bootstrapping algorithm to address missing data that accounts for smooth time trends, changes across cross-sectional variables and time and space correlations, and integrates scant knowledge to specific cells when available (Honaker and King 2010). This form of multiple imputations is used because addressing missing data with listwise deletion or mean substitution may foster the risk of obtaining biased estimates, increasing type II errors, and underestimating correlations and coefficient weights.[13] For a model that combines *assets*, *revenue*, and *employees*, we expand using Chebyshev polynomials because they can arguably approximate most functional forms (Kolsarici and Vakratsas 2015). We opt for a third-order polynomial based on the Bayesian information criterion after considering expansions up to the 20th degree.

Using our base estimation model, we examine the overall MNE donations–economic importance elasticities of the various operationalizations of economic importance and find that it is greater for the ownership-weighted proportion of MNE affiliates in a country than for the alternative measures. Dividing the ratio of the coefficient of the independent variable by the linear prediction results in a higher overall prediction effect. The analyses suggest that economic importance based on the proportion of ownership-weighted MNE affiliates is the best linear unbiased estimator. Thus, we use it as our baseline measure.

Of note, our measure offers a conservative estimate of the economic importance of a country to the MNE in two ways. First, MNEs with no ongoing operations that plan to expand to the country soon may have a comparatively high incentive to donate. Alternatively, firms that plan to exit the country may have a reduced incentive to donate. Given that we calculate this variable on the year of the disruption, the donation of firms with these two strategies should reduce the statistical significance of our predictor. Second, this variable does not include sporadic commercial activity. Consequently, the number of donors with some economic activity with the affected country is likely higher than what the analyses reflect.

Moderators. For Hypothesis 2, the *Dominant MNE Index* is a Herfindahl–Hirschman index calculated at the country level as the sum of squares of the proportion of revenues of the largest five firms in the focal country in the year of the institutional disruption. This variable provides a normalized value of firms with large shares of the country market. It ranges from zero to one with larger values indicating more dominant MNEs in the country market. The raw data source is Orbis.

For Hypotheses 3a and 3b, according to the literature on aid and disaster management, (1) the average foreign net official development assistance and official aid and (2) the country's average gross national expenditure explain the availability of foreign and host country resources toward disruptions, respectively (Cavallo et al. 2013, Becerra et al. 2014). When a disruption occurs, a firm may not know precisely how much general funding will be provided before it donates. Yet the firm can form an estimate based on recent history. Therefore, to avoid reverse causality and capture the information available when making the donation decision, we use data from the two years before the event occurs. We construct *Foreign public aid* with the average of net official development assistance and official aid received for disasters in the two years preceding the disruption. *Host-country public aid* is the average value of gross national expenditure in current U.S. dollars in the previous two years divided by the disaster's *economic damage*.

For Hypothesis 4, we use *regulatory quality* from the World Bank WGI.[14] The variable ranges from −2.5 to 2.5 and "captures perceptions of the ability of the government to formulate and implement sound policies and regulations that permit and promote private sector development" (Kaufmann et al. 2011, p. 4).

Controls. In our most stringent specification, we use firm (i.e., MNE), affiliate industry, country-event, and year fixed effects that absorb all time-invariant donor and industry-specific effects and account for path-dependency of aid and year idiosyncrasies via year fixed effects. Country-event fixed effects are included because disruptions often affect different countries, particularly in some regions of the world, such as the Asian Pacific and Central America. Therefore, the shock data are naturally nested by country-event. Estimation vectors contain a battery of time-variant variables that influence company philanthropy. At the firm level, return on assets (*ROA*) and *Tobin's q*, both winsorized at 5%, and logs of *total assets*, *total revenue*, and *number of employees* proxy for performance and size, which may affect the firm's capacity to donate and the existence of disruption-specific resources (Liang and Renneboog 2017, Patten 2008, Ballesteros and Gatignon 2019, Flammer et al. 2019). Research and development intensity (*R&D intensity*) and *advertising and administration expenses* (logged) may determine intangible resources, such as reputation and visibility, that are behind the capacity to accrue rents from philanthropy (Servaes and Tamayo 2013).[15] To address *donor fatigue*, we include the number of MNE donations made within the previous year (Andreoni 2006). Orbis categorizes affiliates' main activities into four categories: retail, service, manufacturing, and wholesale. We include *business-to-consumer*, measured as a binary indicator set equal to one when the MNE's main activity is retail or services, because it may entail a different

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propensity to donate (Marquis and Tilcsik 2016). These variables are lagged by one year.

At the country level, we include logs of *GDP* adjusted for purchasing power parity in current U.S. dollars, *trade* as a percentage of GDP, *population*, percentage of *urban population*, and land *area* (in square kilometers). These variables may skew shock exposure, the functioning of market-oriented institutions, and the availability of aid (Acemoglu et al. 2005, Kahn 2005, Cavallo et al. 2013, Becerra et al. 2015). We account for a measure of *control of corruption* that indexes "perceptions of the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as 'capture' of the state by elites and private interests" (Kaufmann et al. 2011). We lag these variables by one year.

At the event level, we gather data from EM-DAT, UNOCHA FTS, and Swiss Re to control for *disruption type* (e.g., terrorist attacks, epidemics, floods, earthquakes, etc.) because global philanthropy may be biased to some events (Eisensee and Strömberg 2007) and logs of the *number of deaths* and *people affected* (i.e., displaced and injured) and *economic damage* to control for the magnitude of events (Cavallo et al. 2013). The number of shocks in any given year may decrease the likelihood that a disruption receives donations (Eisensee and Strömberg 2007). We include logs of the number of *disruptions in the country* and *worldwide* in the year before the focal disruption. Finally, we control for other *news pressure* that may crowd out aid by using the average of the median number of minutes a news broadcast devotes to the top three news segments in a day over the 40 days after the disaster date (Stromberg 2007).

Our baseline method uses models of the following form:

$$\begin{aligned} ln(1 + donation_{fcd}) = \beta_0 &+ \beta_1(economic\ importance_{fcd}) \\ &+ \beta_2(e.i._{fcd} x\ dominant\ MNE\ index_{cd}) \\ &+ \beta_3(e.i._{fcd} x\ host\text{-}country\ public\ aid_{cd}) \\ &+ \beta_4(e.i._{fcd} x\ foreign\ public\ aid_{cd}) \\ &+ \beta_5(e.i._{fcd} x\ regulatory\ quality_{cd}) \\ &+ \mu_f + \delta_c + \gamma_d + \varepsilon_{fcd}, \end{aligned} \tag{2}$$

where f is firm, c is country, d is event, *e.i.* is economic importance, $\boldsymbol{\mu}$ is a vector of firm, $\boldsymbol{\delta}$ is a vector of country, and $\boldsymbol{\gamma}$ is a vector of event controls; ε_{fcd} is the error term. We apply generalized least squares with random effects (the p-value for the Hausman test is 0.1037).

Results

Table 1 contains descriptive statistics. Panel A in Table 1 presents information at the MNE level. The average company in our sample has 57,695 employees, $82.8 billion in total assets, and $19.9 billion in revenue. Panel B presents country-level data. The average GDP of the affected countries was $740.6 billion. On average, sample countries have a level of regulatory quality of −0.01, close to the midpoint of zero on the scale. Similarly, countries have mean control of corruption of −0.09, which is comparatively close to the midpoint. Panel C presents disruption-level data. The 265 events have mean impacts of 2,105 deaths, 2.0 million people affected, and $5.5 billion in economic damage. The vast majority of these disruptions are caused by floods and storms with earthquakes the next most common event.

Approximately 31.4% of the MNEs in our sample are donors. For each event during our sample period, the average contribution by MNE donors totals $858,000. A large number of firms do not donate to a given event. As shown in Panel D, which contains the MNE-country-event variables, the average donation is $5,000 once we consider nondonors. The high maximum value of $54.0 million and standard deviation of $220,000 indicate that the distribution of donations has a long right tail. Similarly, economic importance has a right-skewed distribution. There are a number of countries where many MNEs are not located, such as Vanuatu or Mali, which receive a zero for economic importance. Thus, the mean is 0.04 with a standard deviation of 0.15 and a maximum of 1.0.

Table 2 contains the theoretical model adding incrementally more demanding sets of control variables and fixed effects. Model 1 shows the estimate of the effect of economic importance with firm-, country, and event-controls. Model 2 displays the results for the full model without controls. We include time-varying controls in Model 3. Model 4 incorporates firm, industry, country-event, and year fixed effects. We discuss the baseline specification findings that have firm, country, and event controls (Model 3).

The coefficient for economic importance remains positive and significant across specifications. These results suggest that the economic importance of the country is an underlying factor for MNE donations in the aftermath of institutional disruptions.[16] Considering a model with no interactions, a one-unit increase in the economic importance of the country for the MNE raises the average donation over 81% (an estimated coefficient of 0.596), which makes the impact economically significant.

The observed magnitude of this association is influenced by two aspects. First, we focus on the largest MNEs at the international level whose operations tend to be highly geographically diversified. A one-unit increase in the economic importance is extraordinary for these firms. Second, our sample includes a large portion of nondonors, which provides stability and robustness to the estimates and makes the elasticity pronounced. This empirical strategy of considering

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Table 1. Descriptive Statistics

Variables	Observations	Mean	Standard Deviation	Minimum	Maximum
Panel A. Multinational Enterprise					
Return on Assets	1,891	4.21	4.14	−16.84	32.05
Tobin's Q	1,891	0.85	0.82	0.01	6.47
Total Assets	1,891	82,800,000	249,300,000	246,034	2,899,000,000
Total Revenue	1,891	19,940,000	30,530,000	1,346,000	471,900,000
Number of Employees	1,891	57,695	83,246	214	2,202,000
R&D Intensity	1,891	2.97	4.14	−0.06	35.56
Advertising & Admin Expenses	1,891	8,603,000	15,730,000	20,275	119,800,000
Business to Consumer	1,891	0.44	0.41	0.00	1.00
Donor Fatigue	1,891	0.06	0.26	0.00	2.69
Panel B. Country					
Dominant MNE Index	129	0.01	0.07	0.00	0.74
Regulatory Quality	129	−0.01	0.97	−2.00	2.00
GDP (PPP current international)	129	740,600,000,000	2,274,000,000,000	198,200,000	18,050,000,000,000
Population	129	50,000,000	170,100,000	52,045	1,351,000,000
Trade (% of GDP)	129	87.20	54.98	0.00	391.00
Urban Population (%)	129	54.10	23.43	10.50	100.00
Control of Corruption	129	−0.09	1.00	−2.00	2.00
Area (km^2) (ln)	129	12.129	2.807	5	17
Disruptions in Country	129	0.74	0.14	0.69	1.50
Panel C. Event					
Host-Country Public Aid	265	2,271,000,000,000	5,307,000,000,000	244,200,000	20,660,000,000,000
Foreign Public Aid	265	336,600,000	526,400,000	0	2,997,000,000
Earthquake	265	0.11	0.32	0.00	1.00
Epidemic	265	0.02	0.15	0.00	1.00
Extreme Temperature	265	0.03	0.18	0.00	1.00
Flood	265	0.40	0.49	0.00	1.00
Forest Fire	265	0.01	0.09	0.00	1.00
Landslide	265	0.01	0.11	0.00	1.00
Mass Movement	265	0.01	0.11	0.00	1.00
Storm	265	0.35	0.48	0.00	1.00
Terrorist Attack	265	0.01	0.11	0.00	1.00
Volcanic Activity	265	0.01	0.09	0.00	1.00
Wildfire	265	0.03	0.17	0.00	1.00
Number of Deaths	265	2,105	16,983	0	222,570
Number of People Affected	265	2,033,000	8,842,000	0	85,000,000
Economic Damage (USD Million)	265	5,515	18,732	0	210,000
Disruptions Worldwide	265	29.06	12.25	14.00	57.00
News Pressure	265	9.24	3.56	0.00	18.50
Panel D. MNE-Country-Event					
Donation	501,115	5,116	219,707	0	54,000,000
Factors for Market (binary)	501,115	0.17	0.37	0.00	1.00
Economic Importance	501,115	0.04	0.15	0.00	1.00

Notes. Each subheading provides the level of aggregation of the data presented in that part of the table. Panel A provides summary statistics for our analyses based on the data set of the 2,000 largest MNEs at the international level across the 2007–2019 period. Panel B summarizes data when the countries were affected by institutional disruptions. Panel C shows the epidemics, natural disasters, and terrorist attacks whose impacts rank at the 99th percentile in the countries that they affected in the 12-year period. Panel D shows the dependent and two independent variables for the full data set. Economic importance is based on the share of affiliates that the MNE has in a country in any given year, adjusted by proportion of ownership. Factors for market represents donations whose reports explicitly state that the MNE is allocating resources to areas instrumental to restore market-oriented institutions functioning. The data in this table are summarized in original units. Please see the text for further details on variable definitions and construction.

a representative sample of firms and including nondonors adds to the literature, given that previous work has limited analyses to donor companies.

Regarding the interactions, in line with Hypothesis 2, the results suggest that the dominant MNE index positively moderates MNE donations for economically connected firms. Country markets have many large firms, on average, in our sample with a mean value of 0.01. Holding everything else constant at the mean level of the interaction, a one-unit increase in the dominant MNE index is associated with 65% larger donations from MNEs for which the affected country is economically important.

We find mixed results for the interaction of public aid and economic importance on MNE philanthropy. On the one hand, we observe a noncomplete crowding-out effect. As hypothesized in Hypothesis 3a, at the means, a 1% increase of host-country aid is

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Table 2. Economic Importance of the Country to the MNE and Donations to Disruptions

	USD Donation Amount (ln)			
Variables	(1)	(2)	(3)	(4)
H1: Economic Importance	0.596***	0.390***	0.330***	0.207***
	(0.047)	(0.071)	(0.070)	(0.075)
H2: EI x Dominant MNE Index		0.335*	0.646***	0.388**
		(0.185)	(0.186)	(0.193)
H3a: EI x Host-Country Public Aid		-0.000***	-0.000***	-0.000**
		(0.000)	(0.000)	(0.000)
H3b: EI x Foreign Public Aid		0.000***	0.000**	0.000**
		(0.000)	(0.000)	(0.000)
H4: EI x Regulatory Quality		0.542***	0.454***	0.438***
		(0.056)	(0.052)	(0.059)
Dominant MNE Index		-0.038**	-0.005	-0.005
		(0.019)	(0.019)	(0.018)
Host-Country Public Aid		0.000***	-0.000***	-0.000
		(0.000)	(0.000)	(0.000)
Foreign Public Aid		-0.000***	0.000***	-0.000***
		(0.000)	(0.000)	(0.000)
Regulatory Quality		-0.014***	-0.003	-0.004
		(0.001)	(0.002)	(0.008)
Firm Controls	YES	NO	YES	YES
Country Controls	YES	NO	YES	YES
Event Controls	YES	NO	YES	YES
Firm, Industry, Country by Event, Year Fixed Effects	NO	NO	NO	YES
Constant	-0.608***	0.050***	-0.597***	0.000
	(0.076)	(0.004)	(0.075)	(0.000)
Observations	501,115	501,115	501,115	501,115
MNEs	1,891	1,891	1,891	1,891
Countries	129	129	129	129
Disruptions	265	265	265	265

Notes: This table reports regression estimates for the relationship between the economic importance of a country for a firm and donations in the aftermath of institutional disruptions. The database includes the largest 2,000 MNEs at the international level. Institutional disruptions are epidemics, natural disasters, and terrorist attacks whose impacts rank at the 99th percentile in the countries that they affected. The period is 2007-2019. Standard errors are clustered by MNE and reported in parentheses.

*** p<0.01, ** p<0.05, * p<0.1.

associated with smaller donations from economically connected MNEs. Contrary to our expectations, however, a 1% increase in foreign public aid is correlated with larger donations.[17] This result could be driven by MNEs perceiving the amount of foreign aid as a signal of the country's resilience. The more resources that the country receives from abroad, the lower the resilience and the greater the perceived need for firm resources. To further delve into this association, we run the analysis by country income group and present the results in the Online Table 8.The results suggest that, when countries do not have sufficient resources for recovery, there is no crowding-out effect for public aid.

For Hypothesis 4, the level of country regulatory quality appears to be a significant factor when MNEs with economically important operations in the country decide to donate after institutional disruptions. The results indicate that the average giving by these firms rises in regulatory quality. At the mean, a one-unit increase in the affected country's *regulatory-quality index* is associated with an increase in donations from economically connected MNEs of about 45% (see Model 2 in Table 2).

We employ alternative measures of economic importance. Table 3 reports the baseline random-effects model with firm, country, and event controls using a measure of economic importance based on *assets* (Model 1), *revenue* (Model 2), and *employees* (Model 3). Model 4 uses the combined expansion on three variables. The results are qualitatively comparable.

Donations Toward Market Factors. We next examine whether economic importance is predictive of donations, explicitly targeting the restoration of factors necessary for market-oriented institutions to function in Table 4. Model 1 includes *economic importance* without interaction effects and incorporates firm, country, and event controls. Model 2 adds the interaction effects, and Model 3 includes firm, industry, country by event, and year fixed effects. The results indicate that holding everything else constant at the mean, a one-unit increase in the economic importance of the country for the MNE

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Table 3. Alternative Measures of Economic Importance of the Country to the MNE and Donations to Disruptions

	USD Donation Amount (ln)			
Variables	(1) Assets	(2) Revenue	(3) Employees	(4) Combined
H1: Economic Importance (Assets)	0.125***			
	(0.013)			
H2: EI (Assets) x Dominant MNE Index	0.176***			
	(0.037)			
H3a: EI (Assets) x Host-Country Public Aid	-0.000***			
	(0.000)			
H3b: EI (Assets) x Foreign Public Aid	0.000*			
	(0.000)			
H4: EI (Assets) x Regulatory Quality	0.169***			
	(0.010)			
H1: Economic Importance (Revenue)		0.119***		
		(0.013)		
H2: EI (Revenue) x Dominant MNE Index		0.135***		
		(0.038)		
H3a: EI (Revenue) x Host-Country Public Aid		-0.000***		
		(0.000)		
H3b: EI (Revenue) x Foreign Public Aid		0.000		
		(0.000)		
H4: EI (Revenue) x Regulatory Quality		0.167***		
		(0.010)		
H1: Economic Importance (Employees)			0.136***	
			(0.013)	
H2: EI (Employees) x Dominant MNE Index			0.140***	
			(0.038)	
H3a: EI (Employees) x Host-Country Public Aid			-0.000***	
			(0.000)	
H3b: EI (Employees) x Foreign Public Aid			0.000*	
			(0.000)	
H4: EI (Employees) x Regulatory Quality			0.177***	
			(0.010)	
H1: Economic Importance (Combined)				0.062***
				(0.006)
H2: EI (Combined) x Dominant MNE Index				0.076***
				(0.019)
H3a: EI (Combined) x Host-Country Public Aid				-0.000***
				(0.000)
H3b: EI (Combined) x Foreign Public Aid				0.000***
				(0.000)
H4: EI (Combined) x Regulatory Quality				0.091***
				(0.005)
Dominant MNE Index	0.174***	0.183***	0.190***	0.169***
	(0.011)	(0.010)	(0.010)	(0.011)
Host-Country Public Aid	-0.000***	-0.000***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)
Foreign Public Aid	0.000***	0.000***	0.000***	0.000***
	(0.000)	(0.000)	(0.000)	(0.000)
Regulatory Quality	0.004	0.004	0.004	0.003
	(0.003)	(0.003)	(0.003)	(0.003)
Firm, Country and Event Controls	YES	YES	YES	YES
Constant	-0.490***	-0.488***	-0.489***	-0.491***
	(0.036)	(0.036)	(0.036)	(0.036)
Observations	501,115	501,115	501,115	501,115
MNEs	1,891	1,891	1,891	1,891
Countries	129	129	129	129
Disruptions	265	265	265	265

Notes: This table reports regression estimates for the relationship between the alternative constructions of economic importance of the country for a firm and donations in the aftermath of disruptions. Each explanatory variable uses the share of affiliates in the affected country adjusted by ownership. The combined measure in Model 4 is calculated with a third-order Chebyshev polynomial expansion on an additional explanatory variable. In Model 1, the explanatory variable is the share of assets; Model 2, the share of revenue; Model 3, the share of employees. Model 4 uses a Chebyshev polynomial expansion on affiliates and the three other predictors. See text for additional variable definitions and constructions. The database includes the largest 2,000 MNEs at the international level. Institutional disruptions are epidemics, natural disasters, and terrorist attacks whose impacts rank at the 99th percentile in the countries that they affected. The period is 2007-2019. Standard errors are clustered by MNE and reported in parentheses.

*** p<0.01, ** p<0.05, * p<0.1.

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Table 4. Economic Importance of the Country to the MNE and Donations Aimed at Restoring Factors Necessary for Market Functioning

	USD Donation Amount (ln)			
Variables	(1)	(2)	(3)	(4)
H1: Economic Importance	0.920***	0.799***	0.790***	0.738***
	(0.046)	(0.020)	(0.028)	(0.029)
H2: EI x Dominant MNE Index		0.335*	0.634***	0.623***
		(0.186)	(0.187)	(0.185)
H3a: EI x Host-Country Public Aid		-0.000***	-0.000***	-0.000*
		(0.000)	(0.000)	(0.000)
H3b: EI x Foreign Public Aid		0.000***	0.000***	0.000**
		(0.000)	(0.000)	(0.000)
H4: EI x Regulatory Quality		0.797***	0.769***	0.714***
		(0.051)	(0.060)	(0.059)
Dominant MNE Index		-0.003	-0.003	0.006
		(0.004)	(0.004)	(0.005)
Host-Country Public Aid		-0.000	-0.000	0.005
		(0.001)	(0.001)	(2.662)
Foreign Public Aid		-0.000***	-0.000***	0.000**
		(0.000)	(0.000)	(0.000)
Regulatory Quality		-0.000	-0.000	-0.006
		(0.001)	(0.001)	(0.002)
Firm Controls	YES	NO	YES	YES
Country Controls	YES	NO	YES	YES
Event Controls	YES	NO	YES	YES
Firm, Industry, Country by Event, Year Fixed Effects	NO	NO	NO	YES
Constant	-0.138***	-0.016***	-0.256***	0.000
	(0.016)	(0.001)	(0.004)	(0.000)
Observations	501,115	501,115	501,115	501,115
MNEs	1,891	1,891	1,891	1,891
Countries	129	129	129	129
Disruptions	265	265	265	265

Notes: This table reports random effects regression estimates for the relationship between the economic importance of a country for a firm and donations to factors that are necessary for market functioning in the aftermath of institutional disruptions. The database includes the largest 2,000 multinational enterprises (MNEs) at the international level. Institutional disruptions are epidemics, natural disasters, and terrorist attacks whose impacts rank at the 99th percentile in the countries that they affected. The period is 2007-2019. Standard errors are clustered by MNE and reported in parentheses.

*** p<0.01, ** p<0.05, * p<0.1.

increases its donations toward country market factors by 120% (Model 3).

Additional Robustness

Matched Sample. One concern is that, given the geographical (and, thus, political and socioeconomic) heterogeneity of countries exposed to disruptions, firms with a similar propensity to donate may self-select to specific institutional environments. This can introduce bias in the estimation because philanthropy and economic importance may be moving in the same direction as unobservable variables (e.g., disruption-risk aversion). To mitigate this risk, we applied *coarsened exact matching* (Iacus et al. 2011) to balance the baseline propensity to donate between the treatment (i.e., firms with at least one affiliate in the affected country) and the control groups (i.e., firms with no affiliates). The matching uses variables that the extant literature has identified as impacting company philanthropy: *ROA*, *Tobin's q*, *total assets*, *total revenue*, *number of employees*, *R&D intensity*, *advertising and administrative expenses*, *consumer orientation*, *and industry*. Although we cannot rule out the existence of all unobservable effects, the results are qualitatively unaffected by this procedure (see Online Appendix Table 10).

Controlling for Omitted Variables and Confounders

Is It the Pursuit of Reputation? A stream of research suggests that decision makers engage in philanthropy to boost their firms' reputation (Exley 2018). Muller and Kräussl (2011), for instance, proxy reputation for social irresponsibility with the number of concerns in the Kinder, Lydenberg, and Domini database and find that negative scores are associated with a relatively high likelihood of donating in the aftermath of Hurricane Katrina.

To explore the possibility that MNEs' pursuit of reputation explains variance in donations, we calculate a *Janis–Fadner coefficient of imbalance*—a widely used variable of media reputation (see Zhang 2016 for a survey). We use natural language processing to quantify the tone of media reports in Factiva,

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mentioning each MNE in our sample in a period of one year before the start date of the disruption. Online Table 11, which also contains the procedural description, shows that *reputation* becomes statistically insignificant when the model includes *economic importance*. Interaction of the two variables, whose effect is sizeable ($p < 0.01$), indicates that the economic importance of the country to the MNE is a precondition for the pursuit of reputation. This suggests that economic importance mediates the effects of the quest for accumulating reputational capital on donating in this context.

As an additional validation, we explore the predictive power of *reputation* on the likelihood of donating toward factors necessary for market-oriented institutions to function. We substitute economic importance with our reputation variable in the specification of Table 2. The coefficient of such a variable is not statistically significant (Online Table 12).

Is It the Pursuit of Market Standing? An established argument in the nonmarket literature is that corporations engage in philanthropy to enhance their market position. The strength of this motive is inversely proportional to country market standing. Therefore, one expectation is that low-standing organizations should donate relatively large amounts because the marginal utility of such strategies is higher for them than for higher standing firms (Servaes and Tamayo 2013, Liang and Renneboog 2017). To address this potential confounder, we proxy *market standing* by the rank of the MNE in the country by firm revenue the year before the disruption. The coefficient of interaction with *economic importance* implies that the donations from economically connected MNEs fall with every standard-deviation decrease in *market standing* (see Online Table 13).

Is It the Pursuit of Political Favors? Another prediction is that firms construct their nonmarket behavior in accordance with signals coming from influential government actors (Zhang and Luo 2013, Marquis and Qian 2014, Bertrand et al. 2020). Under this argument, when the domestic government allocates aid, firms increase their giving to seek its preference and capture political favors for rent-seeking opportunities. The analyses for Hypothesis 3a systematically reject the likelihood of such a motive being a part of the studied donations.

Is It Altruism or Reciprocity? Firms may donate in response to social preferences from internal stakeholders. For instance, the reciprocal motives of employees may foster the philanthropic behavior of MNEs (Charness and Rabin 2002, Fehr and Fischbacher 2002, Small and Simonsohn 2008). The risk of this confounder is essential in our setting given that research shows that people react more strongly to shock-related losses than to chronic conditions (Small 2010, Ballesteros and Pamphile 2021). We find that the interaction of *employee-driven donation* with *economic importance* is statistically significant and negative (Online Table 14), suggesting that, when MNEs donated following an initiative by employees, the donation amount was significantly lower than when not.

To further test the argument that business decision makers donate to satisfy altruistic motives, we incorporate a ratio of *human to economic loss*, operationalized as the number of deaths divided by economic damage. The measure captures the relative humanitarian damage. Results indicate that the donation amount decreases in the interaction of *human to economic loss* with *economic importance* (Online Table 15). The result is consistent with the skewness of donations to institutional disruptions with substantial economic costs vis-à-vis human impacts across the events for 1997–2019 as documented in the online appendix.

Is It Media Salience? Studies show that media can influence humanitarian aid, but its influence is heterogeneous across events and skewed to some types of disasters (e.g., earthquakes) and countries (e.g., higher income) (Eisensee and Strömberg 2007, Franks 2013). For some institutional disruptions, economic importance may be correlated with countries that receive a sizeable magnitude of news coverage. Events such as the 2017 Hurricane Maria in Puerto Rico and the 2011 earthquake and tsunami in Japan were certain to receive firm aid because of media coverage. We, thus, follow Stromberg (2007) and run analyses with only disruptions that had a probability of being in the news of 50% and lower. The magnitude and direction of the estimates hold (Online Table 16).

Is It Social Pressure Coming from Local Presence? Firms are embedded in societal arrangements that foster normative pressure for philanthropy on the geographically proximate firm (Marquis et al. 2013, Zhang and Luo 2013) and may donate for the approval of the local community (i.e., social license to operate; Boutilier and Thomson 2011, Wilburn and Wilburn 2011). The MNE's physical presence may be a simpler explanation, and economic importance would be a second order measure captured by differentiating firms based on geographic presence. We test this by substituting our predictor with *physical presence*, a binary variable taking value one when the firm has at least one affiliate in the country. Online Table 17 shows that its coefficient is negative and significant, consistent with work suggesting that firms present in a country struck by a major disaster engage in scarce giving (Tilcsik and Marquis 2013). Given that physical

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presence weights all locations equally, this result reconciles with existing work and, thus, implies that not considering the strategic value of affiliates is econometrically important. It suggests that the mechanism driving MNE philanthropy in the aftermath of institutional disruptions is more complex than local presence alone, and our measure captures such complexity.

Is It Inequity or Poverty Aversion? Firms may allocate resources to reduce financial disparity (Camerer and Fehr 2005, Fehr et al. 2006). If this preference dominates restoring market-oriented institutions, MNE philanthropy would be comparatively greater in countries with high poverty levels in which economic inequality is pervasive. To test this argument, we regress an interaction of *economic importance* and *poverty* using the poverty headcount ratio at $3.20 a day (2011 PPP) (as a percentage of population) from the WDI (see Anand and Sen 2000). We find that MNEs in the sample donated in a lower magnitude to more impoverished countries than to higher-income countries (Online Table 18).

Discussion and Conclusion

This study investigates how the economic importance of a country to an MNE spurs its philanthropy in the aftermath of institutional disruptions. Situated at the intersection of scholarly work on the drivers of organizational nonmarket behavior and on organizational response to the institutional environment (Aguilera and Grøgaard 2019; Ahuja and Yayavaram 2011; Bonardi 2004; Doh et al. 2017), this paper responds to calls for research that integrates these independent literatures (Dorobantu et al. 2017).

Contribution to the Nonmarket Literature on Company Philanthropy

We theorize that company philanthropy in the aftermath of institutional disruptions is driven by a motive not previously studied: restoring the country market. Firms with operations in an affected country are likely to incur substantial damages. Therefore, extant studies predict that these firms will engage in scarce giving after large shocks. We replicate the results of extant work and find a negative relationship with physical presence. However, once we consider the economic importance of firm locations, we find that MNEs facing the prospect of performance shocks resulting from the economic importance of the affected country to the firm allocate resources to help its recovery.

Prior work on company philanthropy emphasizes that donors pursue firm-specific benefits, such as reputation and political favors. Our study introduces a strategic motive whose benefits approximate a public good (i.e., it is not firm-specific). That is, it can be difficult to disentangle private benefits from those received by other organizations and society at large. Given its public nature, this type of philanthropy is more susceptible to collective action issues. For instance, some companies free ride on the contribution of others and still benefit from the outcome. Consequently, a measure of economic importance facilitates the identification of the pool of firms that are likely to become substantial contributors to the recovery of the country market.

A key implication is that our moderator predictions differ from those in the extant literature. For instance, a prevailing logic is that firms in competitive country markets have greater incentives to engage in philanthropy because the potential returns to firm differentiation are larger when there are fewer dominant firms (Liang and Renneboog 2017). Accordingly, if there is demand for nonmarket behavior, firms engage in such behavior with the hope of achieving or sustaining a competitive advantage (Kaul and Luo 2017). Conversely, our findings show that companies in highly competitive country markets tend to donate comparatively less because other firms can contribute.

Similarly, studies predict that firms use philanthropy to foster stakeholder relationships by aligning their giving with the interests of influential host country actors and generate firm political favors (Bertrand et al. 2020), particularly when such giving occurs in countries with low institutional development (Zhang and Luo 2013, Hornstein and Zhao 2018). In contrast, we find that firms give less when the host country can provide resources for recovery and find a strong positive link between donation magnitudes and regulatory quality. The results underscore the importance of the market restoration motive after institutional disruptions as firm behavior differs from commonly studied behaviors under stable contexts.

More broadly, the findings suggest that the motives of pursuing reputation, meeting social pressures, achieving market standing or media attention, or satisfying reciprocal and altruistic goals do not replicate the explanatory power of the economic importance of a country to the MNE after institutional disruptions. Although company philanthropy toward institutional disruptions may still result in a warm glow or rents via a signal of product quality and willingness to pay, the robustness analyses indicate that the economic importance of restoring country markets may mediate strategic considerations and social preferences.

Our study connects with theories on the importance of communities for "organizations in a global age" (Marquis and Battilana 2009, p. 283). Although the existing literature considers how firms imitate other organizations in their community (Marquis and Tilcsik 2016) and how the social embeddedness of headquarters and other links with local stakeholders foster

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social and political pressures that activate firm giving (Luo and Chung 2013), it has not explored the effect of differences in the economic importance of the geographic location to the firm. Our analyses show that such a measure can help illuminate the conditions and mechanisms under which firms with a worldwide presence respond to local causes.

Finally, this study adds to the literature on international nonmarket strategy. Traditionally, company giving is evaluated within countries or with a few single-country events, which has constrained our understanding of how differences in the institutional environment affect firm engagement in societal issues. The longitudinal data set alleviates the risks of measurement error and omitted-variable bias that have been a concern for observational studies (Liang and Renneboog 2017).

Contribution to the Literature on Organizational Responses to the Institutional Environment

We also contribute to the theoretical and empirical literature that examines firms' response to the institutional environment (see, e.g., Khanna and Palepu 1997, Peng 2003, Ahuja and Yayavaram 2011, Dutt et al. 2016, Doh et al. 2017, Aguilera and Grøgaard 2019). We formalize the concept of institutional disruptions based on the four characteristics of temporality, suddenness and unexpectedness, severity, and pervasiveness. Despite extensive research on the enduring absence of country market-based institutions (i.e., voids) and evolutionary institutional changes, the concept of institutional disruptions has yet to be fully developed. Systematic attention to this context is particularly pertinent to the study of MNEs that are increasingly exposed to a wide range of disruptions vis-à-vis domestic firms. Interestingly, the country markets that have experienced substantial growth in the last three decades, such as China and India, also top the list in terms of exposure to large shocks. The characteristics of disruptions are key to understanding why MNEs increasingly allocate resources toward country market restoration. Considering the trends of globalization and economic integration, the strategic value of these country markets for international business makes exiting or avoiding shock-prone locations an economically unattractive option.

The findings offer a fresh perspective of how organizations affect institutional development. A large literature studies how organizations may seek to influence public policy for market-oriented institutions that benefit the organization at the expense of social welfare (Ahuja and Yayavaram 2011, Bertrand et al. 2020). We propose that MNEs donate their resources and, therefore, influence institutions in a fundamentally different way. Firm action toward country market recovery may have substantial positive implications (High-Level Panel on Humanitarian Financing 2016, Ballesteros et al. 2017, Ballesteros and Gatignon 2019). Donations to restore country market–oriented institutions are the opposite of starvation strategies used by firms to constrain the resources for such institutions by directly providing resources to rebuild them. We offer an empirical test for theoretical work on when firms use a nonmarket strategy that creates a social benefit (Douty 1972, De Alessi 1975, Kaul and Luo 2017).

This study points to the importance of the quality of country market–oriented institutions for garnering firm donations for recovery. The results of our analysis by country income level suggests that regulatory quality is crucial in lower income countries for economically connected MNEs to donate. When economically connected firms are faced with greater inefficiencies, bureaucracy, and logistical complexity, their incentives to donate are tempered. For these countries, an implication of our results is that augmenting the investment in structures to manage firm response may facilitate greater support when disruptions strike.

Our findings suggest that institutional disruptions are a context in which the goals of some organizations are more closely aligned with society. For instance, the results pose an interesting paradox on the economic value of monopolistic firms. A vast literature suggests that firms with large shares of the country market seek to influence institutions to enhance their profits, maintain voids, and restrict competition. However, our theory suggests that monopolistic firms have substantial incentives to contribute after disruptions. Altogether, the findings indicate that large MNEs are critical for the country during institutional disruptions that overwhelm the financial and operational capacity of governments.

Limitations and Future Research

Important limitations and boundary conditions remain. First, we restrict the analyses to large, publicly traded firms. Although this type of organization accounts for nearly 90% of the recorded corporate donations over the observed period (and, thus, selection bias is mitigated), future research may explore whether the philanthropy of smaller or private firms may follow different mechanisms. Second, our theory and empirics are at the country level, whereas some disruptions may be regional. Our method of identifying large shocks with systemic effects at the 99th percentile of severity increases the construct validity of institutional disruptions. Although our econometric analysis provides a conservative approach, a promising avenue of investigation is the use of more fine-grained data. Third, our measure of economic importance does not include exports and imports and sporadic economic activity not represented by company affiliates. Our findings, thus, underestimate

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the effect of economic importance. Future consideration of intrafirm and interfirm trade can provide meaningful insights on international interdependencies.

Practical Contributions

Through influencing economic activity, institutions affect the welfare of the constituents in society (Coase 1960) and have often been viewed as central causes and/or solutions to societal issues (Mair and Marti 2009, Vakili and McGahan 2016). Given their tremendous economic impact (Barro 2009, Baker and Bloom 2020), the role of MNEs in responding to institutional disruptions has important implications for society. During the World Humanitarian Summit in 2016, the United Nations highlighted the need to formalize firm participation to help close the growing financial gap of disaster costs (High-Level Panel on Humanitarian Financing 2016). Public aid has been unable to keep pace with the economic costs of shocks that have increased sixfold since 1990. We hope that our study can help government and multilateral agencies to predict the outpouring from firms toward relief and recovery. Collectively, our findings inform on the contextual determinants that affect the MNE incentives underlying this action. This can help stimulate and coordinate the engagement of firms for overcoming challenges from disruptions worldwide.

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Endnotes

[1] We also refer to high economic importance of the affected country to the MNE as "economically connected MNEs."

[2] We conducted 148 unstructured interviews with business decision makers, such as chief financial officers, risk managers, and other employees from 102 S&P 500 firms on their firms' strategies and actions dealing with disasters and terrorist attacks and their consequences. The interviews occurred between 2012–2017 and covered a broad range of MNEs across industry sector and size (see the online appendix at https://institutionaldisruptions.com/ for details).

[3] In the economics literature on shocks, scholars study uncertainty shocks based on volatility and describe their systemic and temporal nature (Bloom 2009, Aghion et al. 2021, Kozeniauskas et al. 2018). However, uncertainty shocks can differ in their severity and the extent to which they are expected. Other scholars have differentiated shocks based on their unexpectedness and the magnitude of their impact (see, e.g., Barro 2009), but these shocks may also have lasting impact rather than temporal effects.

[4] Punctuated changes may result in preexisting market-oriented institutions disappearing, such as that experienced in Central and Eastern European countries during the upheaval from communistic to market-based economies, but they ultimately result in a radically new, permanent institutional state (Newman 2000).

[5] Lobbying for market-oriented institutions often take years to achieve outcomes (Hillman and Keim 1995).

[6] Conversely, firms without economically important operations in the affected country may be motivated to donate for the satisfaction of social preferences, such as altruism, reciprocity, and homophily. Additionally, large shocks frequently receive substantial global media attention, which offers the prospect, for instance, to foster a public image. To illustrate, the 2010 Haitian earthquake was covered extensively in the media. Without Haiti being economically important, worldwide media coverage creates a global opportunity for firms to generate reputational capital.

[7] Host country public aid primarily is sourced from the national governments and allocated to multilateral agencies or to local governments, which, in turn, distribute the funds. Considering institutional disruptions are based on the most severe, high-magnitude shocks, we expect the national government to intervene and be the primary contributor.

[8] Less than 1% of the donations in the sample required converting goods to donation market values.

[9] The WDI database contains internationally compatible statistics on 1,600 time-series indicators for 217 economies and more than 40 country groups. The database is compiled by the World Bank.

[10] Events included in the EM-DAT must meet at least one of the following criteria: 10 or more people killed, 100 or more people affected, a declaration of a state of emergency, or a call for international assistance. Further information can be accessed at http://www.emdat.be/.

[11] Despite being the most used disaster database in empirical studies, EM-DAT has pervasive data inaccuracies and missingness. A substantial part of our work was dedicated to collecting disruptions, and we received critical assistance from UNOCHA and Swiss Re. We conducted independent checks for data accuracy (see the online appendix at https://institutionaldisruptions.com/).

[12] See https://datahelpdesk.worldbank.org/knowledgebase/articles/378833-how-are-the-income-group-thresholds-determined.

[13] A review of how this method can produce more accurate imputation, particularly for data used in the social sciences, than traditional procedures can be found in Blackwell et al. (2017).

[14] According to the World Bank, the WGI is a data set summarizing the views on the quality of governance provided by a large number of enterprise, citizen, and expert survey respondents in industrial and developing countries. The six broad dimensions of WGI governance are rule of law, voice and accountability, political stability and absence of violence, government effectiveness, regulatory quality, and control of corruption (Kaufmann et al. 2011).

[15] Orbis does not separate advertising and marketing from general and administrative expenditures. Therefore, we relied on the full measure of advertising and general and administrative expenses for our control variable.

[16] We apply the Freedman–Lane semipartialing method and find smaller standard errors than linear models, indicating that our main estimation is conservative.

[17] The estimates in Table 2 rely on the nonlogged values of public aid. Using logged values, at the means, a 1% increase of host

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country aid is associated with 25.4% smaller donations for economically connected MNEs. A 1% increase in logged foreign public aid is correlated with 1.4% larger donations from MNEs for which the country market is economically important.

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